

05_gemini_cleanup

March 24, 2026

1 05 Gemini Cleanup and Evaluation

This notebook demonstrates using the Google Gemini API as a late-stage post-OCR correction step. It compares CER/WER before and after cleanup, showing the delta contributed by the LLM correction.

The Gemini prompt is specifically tuned for 17th-century printed Spanish text: it corrects obvious OCR mistakes without modernizing authentic period spellings.

```
[ ]: import os, sys
      from pathlib import Path

      cwd = Path.cwd().resolve()
      ROOT = cwd if (cwd / 'src').exists() else cwd.parent
      if str(ROOT) not in sys.path:
          sys.path.insert(0, str(ROOT))

      # Set your API key here or export GEMINI_API_KEY in your shell
      os.environ.setdefault('GEMINI_API_KEY', 'OUR_GEMINI_API_KEY')

      from src.postprocess_llm import GeminiCleaner, RuleBasedCleaner
      from src.utils import CleanupConfig, read_jsonl
      from src.evaluate import compute_cer, compute_wer
```

1.1 1. Test the Gemini connection

```
[2]: config = CleanupConfig(backend='gemini', gemini_model='models/
      ↪gemini-flash-lite-latest')
      gemini = GeminiCleaner(config)
      print(f'Gemini available: {gemini.is_available}')

      if gemini.is_available:
          test_text = 'Io, P. GARCIA de Ia COMP. de JESUS'
          cleaned, backend = gemini.clean(test_text)
          print(f'Raw:      {test_text}')
          print(f'Cleaned: {cleaned}')
          print(f'Backend: {backend}')
```

Gemini available: True
Raw: Io, P. GARCIA de Ia COMP. de JESUS
Cleaned: Io. P. GARCIA de la COMP. de JESUS
Backend: gemini

1.2 2. Compare rule-based vs Gemini cleanup on baseline predictions

Load the predictions file written by run_baseline.py and re-run both cleaners.

```
[ ]: PREDICTIONS_FILE = ROOT / 'data' / 'predictions' / 'baseline_predictions.jsonl'
GROUND_TRUTH_FILE = ROOT / 'data' / 'ground_truth' / 'ground_truth.jsonl'

if not PREDICTIONS_FILE.exists():
    print('No predictions file. Run run_baseline.py first.')
else:
    predictions = read_jsonl(PREDICTIONS_FILE)
    ground_truth = {r['page_id']: r['text'] for r in
↳read_jsonl(GROUND_TRUTH_FILE)}
    rule_cleaner = RuleBasedCleaner(config)

    results = []
    for pred in predictions:
        page_id = pred['page_id']
        if page_id not in ground_truth:
            continue
        ref = ground_truth[page_id]
        raw = pred.get('raw_ocr', '')
        rule_cleaned = rule_cleaner.clean(raw)

        if gemini.is_available:
            gemini_cleaned, _ = gemini.clean(raw)
        else:
            gemini_cleaned = rule_cleaned

        results.append({
            'page_id': page_id,
            'raw_cer': compute_cer(ref, raw),
            'rule_cer': compute_cer(ref, rule_cleaned),
            'gemini_cer': compute_cer(ref, gemini_cleaned),
            'raw_wer': compute_wer(ref, raw),
            'rule_wer': compute_wer(ref, rule_cleaned),
            'gemini_wer': compute_wer(ref, gemini_cleaned),
        })
        print(f"{page_id}: raw_CER={results[-1]['raw_cer']:.4f}  ↳
↳rule_CER={results[-1]['rule_cer']:.4f}  ↳
↳gemini_CER={results[-1]['gemini_cer']:.4f}")

    if results:
```

```

    avg = lambda key: sum(r[key] for r in results) / len(results)
    print(f"\nAverage CER - Raw: {avg('raw_cer'):.4f} Rule:␣
↪{avg('rule_cer'):.4f} Gemini: {avg('gemini_cer'):.4f}")
    print(f"Average WER - Raw: {avg('raw_wer'):.4f} Rule: {avg('rule_wer')}:
↪.4f} Gemini: {avg('gemini_wer'):.4f}")

```

```

buendia_instruccion_page_0001: raw_CER=0.9480 rule_CER=0.9463
gemini_CER=0.8650
buendia_instruccion_page_0002: raw_CER=0.9181 rule_CER=0.9209
gemini_CER=0.9192
buendia_instruccion_page_0003: raw_CER=1.5745 rule_CER=1.5597
gemini_CER=1.5236

```

```

Average CER - Raw: 1.1468 Rule: 1.1423 Gemini: 1.1026
Average WER - Raw: 1.3387 Rule: 1.3050 Gemini: 1.2879

```

1.3 3. Side-by-side example

Pick a single page and display the raw OCR, rule-based cleanup, and Gemini cleanup side by side.

```

[4]: if results:
    # Use the page with the highest raw CER for a dramatic comparison
    worst = max(results, key=lambda r: r['raw_cer'])
    pred = next(p for p in predictions if p['page_id'] == worst['page_id'])
    ref = ground_truth[worst['page_id']]
    raw = pred.get('raw_ocr', '')

    print(f"=== Page: {worst['page_id']} ===")
    print(f"Raw CER {worst['raw_cer']:.4f} -> Gemini CER {worst['gemini_cer']}:
↪.4f}")
    print()
    print('--- GROUND TRUTH (first 400 chars) ---')
    print(ref[:400])
    print()
    print('--- RAW OCR ---')
    print(raw[:400])
    print()
    gemini_cleaned, _ = gemini.clean(raw) if gemini.is_available else␣
↪(rule_cleaner.clean(raw), 'rule')
    print('--- GEMINI CLEANED ---')
    print(gemini_cleaned[:400])

```

```

=== Page: buendia_instruccion_page_0003 ===
Raw CER 1.5745 -> Gemini CER 1.5236

```

```

--- GROUND TRUTH (first 400 chars) ---
crianza de la niñez. Assi sea,
Divinissimo Niño, por vuestra

```

gracia, assi sea, a vuestra mayor gloria. Amen.

CENSURA DEL R. P. ANTONIO CO-
dorniu de la Compañía de Jesus, Maestro que fue de Theologia, Examinador Synodal de los Obispos de Gerona, Urgel, y Barcelona, Oc.
DE orden del Ilustre Señor Don Francisco Drechos, Canonigo, y Sacristan Dignidad de la Santa Iglesia de Gerona, y Vicari

--- RAW OCR ---

WWW.LLM

GARDORIZIO ON 6020 : LAKEL# : 1 1 1
GOOD DARACON DIOS ON IA DEVORA LAVINAS BENDICONES, LES UNILLEIS.
MAESTRIA LOS TEMPLOSJA PEDAD A VEGERO BARADO PECHO ONA-
COM LOS PADIES EN LA OBEDIENIA, WILSIMOS ADRAOS, DICHOTA EDAD, M
ITEMS INCLUDS; #2 INDOCONA, F OR# QPC OS MOTOCO GAN REGALADOS CR- "
TEO DE LABER Y COM IOS MAYORES, MISS!
ENTANGO MAS OF OUR, I PERSON. Y DOES ON A ORIGINAL IETA-
NET TOT

--- GEMINI CLEANED ---

GARDORIZIO ON 6020 : LAKEL# : 1 1 1
BUEN DARACON DIOS ON IA DEVORA LAVINAS BENDICONES, LES UNILLEIS.
MAESTRIA LOS TEMPLOS JA PEDAD A VEGERO BARADO PECHO ONA-
CON LOS PADRES EN LA OBEDIENCIA, WILISIMOS ADRAOS, DICHA EDAD, M
ITEMS INCLUIDOS; #2 INDOCONA, POR# QVE OS MOSTRÓ GAN REGALADOS CR-"
TEO DE LABOR Y COMIO MAYORES, MISS!
ENTANGO MAS OF OUR, I PERSONA Y DEOS ON A ORIGINAL IETA-
NET TOTAL AMOUNT